

Mathematics for Computing

Assignment

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Introduction to Statistics

Statistics is the science of **collecting, organizing, analyzing, interpreting, and presenting data**. It helps us understand large amounts of information and make better decisions.

For example, statistics can be used to study **population, education, employment, health, business, and the economy**. Governments and organizations collect data, organize it using tables and graphs, analyze it using statistical methods, and use the results for planning and decision-making.

In simple words:

Data Collection → Organization → Analysis → Interpretation → Decision Making

Thus, statistics helps us convert **raw data into useful information**.

Types of Statistics

Statistics is mainly divided into two types:



1. Descriptive Statistics

Used to collect, organize, summarize, and present data.

Examples: Mean, Median, Mode, Variance, Tables, Graphs, and Histograms.

2. Inferential Statistics

Used to make conclusions or predictions about a population using a sample.

Examples: Sampling, Confidence Intervals, Hypothesis Testing, Estimation, and Prediction.

- The **Department of Economics and Statistics (DES)** is the nodal agency for official statistics in Kerala.
- Collects, compiles, analyzes, interprets, and publishes statistical information.
- Collects data related to:
 - Agriculture, Industry, Employment, Prices, Population
- Prepares important economic indicators such as:
 - Gross State Domestic Product (GSDP)
 - Per-capita Income
- Collects farm, wholesale, and retail price data and prepares price indices.
- Conducts surveys and studies for government planning, policy-making, and evaluation of development programmes.
- Publishes statistical reports, databases, and analytical information for government agencies, researchers, and the public.

- **MoSPI** coordinates India's national statistical system and statistical activities.
- Works with Central Ministries, State Governments, and Union Territories.
- Sets standards for data collection, processing, analysis, and dissemination.
- Prepares important national statistics, including National Accounts and national income estimates.
- Conducts surveys on employment, household consumption, enterprises, and other socio-economic areas.
- Develops indicators for monitoring the **Sustainable Development Goals (SDGs)**.
- Monitors major government projects and programmes, including **MPLADS**.
- Supports planning, policy-making, and national development.

Types of Data-Univariate Data

Definition:

Univariate data contains information about only **one variable**.

Example: Student Marks

$$X = \{60, 70, 75, 80, 90\}$$

Here, only one variable, **marks**, is studied.

Application:

Univariate data can be analyzed using measures such as:

- Mean
- Median
- Mode
- Variance
- Histogram

Types of Data-Bivariate Data

Definition:

Bivariate data contains information about **two variables**. It is often used to study the relationship between the variables.

Example: Height and Weight

$$(X, Y) = \{(160, 55), (165, 60), (170, 65)\}$$

where:

- $X = \text{Height}$
- $Y = \text{Weight}$

Application:

Bivariate data can be used to study relationships using:

- Scatter plots
- Correlation
- Regression

Types of Data-Multivariate Data

Definition:

Multivariate data contains information about **more than two variables**.

Example: Student Information

(Age, Height, Weight, Marks)

Here, several variables are studied together.

Application:

Multivariate data analysis is useful for studying complex data where many variables influence each other.

- Machine Learning
- Data Science
- Prediction and Classification

Measures of Central Tendency

Definition:

Measures of central tendency are statistical measures used to find the **central or typical value** of a dataset.

The three main measures are:

- **Mean:** The average value of the data.
- **Median:** The middle value when data is arranged in order.
- **Mode:** The value that occurs most frequently.

Example:

2, 4, 4, 6, 8

$$\text{Mean} = \frac{2 + 4 + 4 + 6 + 8}{5} = 4.8 \quad \text{Median} = 4 \quad \text{Mode} = 4$$

In simple words: These measures help us find a single value that represents the center of the data.

Measures of Dispersion

Definition:

Measures of dispersion show how **spread out or scattered** the values are in a dataset.

Main Measures of Dispersion:

- **Range:** Difference between the largest and smallest values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

- **Variance:** Measures how far the data values are spread from the mean.
- **Standard Deviation:** Shows the amount by which data values differ from the mean.
- **Interquartile Range (IQR):** Measures the spread of the middle 50% of the data.

$$\text{IQR} = Q_3 - Q_1$$

Example:

2, 4, 6, 8, 10

$$\text{Range} = 10 - 2 = 8$$

Measures of Position

Definition:

Measures of position show the **relative location of a value** within a dataset.

Main Measures:

- **Quartiles:** Divide the data into four equal parts.
- **Deciles:** Divide the data into ten equal parts.
- **Percentiles:** Divide the data into one hundred equal parts.

Example:

If a student is in the 90th percentile, the student's score is higher than approximately 90% of the scores.

Application:

Measures of position are used to compare the relative standing of values in a dataset.

Interquartile Range (IQR)

Definition:

The **Interquartile Range (IQR)** measures the spread of the **middle 50% of the data**.

Formula:

$$\text{IQR} = Q_3 - Q_1$$

where:

- Q_1 = First Quartile (25th percentile)
- Q_3 = Third Quartile (75th percentile)

Example:

Consider the ordered data:

2, 4, 6, 8, 10, 12, 14, 16

$$Q_1 = 5, \quad Q_3 = 13$$

Therefore:

$$\text{IQR} = Q_3 - Q_1 = 13 - 5 = 8$$

In simple words: IQR tells us how widely the middle half of the data is spread.

Outliers

Definition:

An **outlier** is a data value that is very different from most of the other values in a dataset.

IQR Method for Detecting Outliers:

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Lower Limit} = Q_1 - 1.5(\text{IQR})$$

$$\text{Upper Limit} = Q_3 + 1.5(\text{IQR})$$

Any value:

$$< \text{Lower Limit} \quad \text{or} \quad > \text{Upper Limit}$$

is considered an **outlier**. **Example:**

$$2, 4, 6, 8, 10, 12, 14, 30$$

The value **30** is far from most other values and may be considered an outlier.

In simple words: An outlier is an unusual value that is far away from most of the data.

Definition:

Skewness is a measure of the **asymmetry** of a data distribution.

Types of Skewness:

- **Zero Skewness:** The distribution is symmetric.
- **Positive Skewness:** The distribution has a longer tail on the right side.
- **Negative Skewness:** The distribution has a longer tail on the left side.

Relationship between Mean, Median and Mode:

Negative Skew: Mean $\neq$ Median $\neq$ Mode

Symmetric: Mean = Median = Mode

Positive Skew: Mode $\neq$ Median $\neq$ Mean

Application: Skewness helps us understand the shape and asymmetry of data.

Definition:

Kurtosis is a measure that describes the **shape, peakedness, and tail behavior** of a data distribution.

Types of Kurtosis:

- **Mesokurtic:** A distribution with normal or moderate peakedness.
- **Leptokurtic:** A distribution with a sharper peak and heavier tails.
- **Platykurtic:** A distribution with a flatter peak and lighter tails.

Kurtosis Value:

$\text{Kurtosis} = 3 \Rightarrow \text{Mesokurtic}$

$\text{Kurtosis} > 3 \Rightarrow \text{Leptokurtic}$

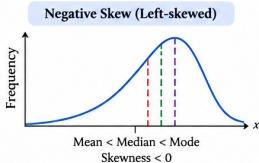
$\text{Kurtosis} < 3 \Rightarrow \text{Platykurtic}$

Application: Kurtosis helps us understand the shape and tail behavior of a data distribution.

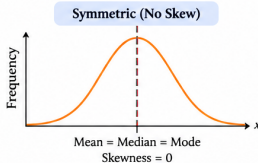
Graphical Representation of Skewness and Kurtosis

Skewness (Asymmetry of a Distribution)

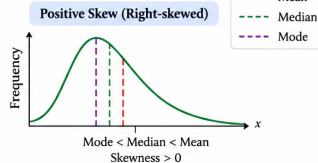
Skewness shows the degree and direction of asymmetry in a distribution.



- Long tail on the left
- More low values



- Perfectly symmetric
- Left and right sides are mirror images

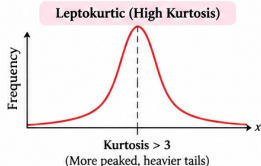


- Long tail on the right
- More high values

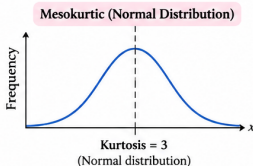
--- Mean
--- Median
--- Mode

Kurtosis (Peakedness of a Distribution)

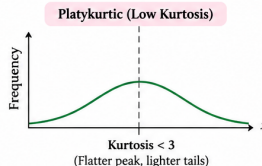
Kurtosis shows how peaked or flat a distribution is compared to the normal distribution.



- Tall and narrow peak
- Heavier tails (more outliers)



- Moderate peak
- Tails as in normal distribution



- Flat and wide peak
- Lighter tails (fewer outliers)

Histogram

A histogram is a graphical representation used to understand the distribution of numerical data.

- Numerical data is divided into intervals called **bins**.
- The height of each bar represents the frequency of observations.
- The bars are placed next to each other without gaps.
- For unequal class intervals, frequency density is calculated as:

$$\text{Frequency Density} = \frac{\text{Frequency}}{\text{Class Width}}$$

- It helps us understand the shape, center, and spread of data.

Sample Mean

Definition

Sample mean is the average value of all observations in a sample. It represents the central value of the given data.

Equation

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where:

- x_i = individual observation
- n = total number of observations
- $\bar{x}$ = sample mean

Example For the data 2, 4, 6, 8:

$$\bar{x} = \frac{2 + 4 + 6 + 8}{4} = 5$$

Application: Used in data analysis to find the average value of a dataset.

Sample Variance

Definition: Sample variance measures how much data values are spread around the sample mean.

Equation:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where x_i is an observation, $\bar{x}$ is the mean, and n is the sample size.

Example:

For 2, 4, 6, where $\bar{x} = 4$:

$$s^2 = \frac{(2-4)^2 + (4-4)^2 + (6-4)^2}{3-1} = 4$$

Application: Used to measure the variability or consistency of data.

Definition: Order statistics are the observations of a sample arranged in ascending or descending order.

Notation:

For a sample $X_1, X_2, \dots, X_n$, after arranging the values:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

where $X_{(1)}$ is the smallest value and $X_{(n)}$ is the largest value.

Example:

For the data 5, 2, 8, 1, 6:

$$1, 2, 5, 6, 8$$

Thus,

$$X_{(1)} = 1, \quad X_{(5)} = 8$$

Application: Used to find the minimum, maximum, median, percentiles, and other ranked values in data.

Sample Covariance

Definition:

Sample covariance measures how two variables change together.

- Positive covariance: Both variables tend to increase or decrease together.
- Negative covariance: One variable tends to increase while the other decreases.
- Zero covariance: No linear relationship is indicated.

Equation:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

where X_i and Y_i are observations, and $\bar{X}$ and $\bar{Y}$ are their sample means.

Application: Used to study relationships between variables, such as study hours and examination marks.

Sample Covariance: Example

Consider two students:

$$X = (1, 3), \quad Y = (2, 6)$$

Step 1: Find the means

$$\bar{X} = \frac{1+3}{2} = 2, \quad \bar{Y} = \frac{2+6}{2} = 4$$

Step 2: Calculate covariance

$$\begin{aligned} s_{XY} &= \frac{(1-2)(2-4) + (3-2)(6-4)}{2-1} \\ &= \frac{2+2}{1} = 4 \end{aligned}$$

Result: Covariance is positive, so X and Y increase together.

Sample Covariance Matrix

Definition:

A sample covariance matrix shows the variance and covariance between two or more variables.

Equation:

For two variables X and Y :

$$S = \begin{bmatrix} s_X^2 & s_{XY} \\ s_{XY} & s_Y^2 \end{bmatrix}$$

where:

- s_X^2 = variance of X
- s_Y^2 = variance of Y
- s_{XY} = covariance between X and Y

Application:

Used in machine learning and data analysis to understand relationships between multiple variables.

Sample Covariance Matrix: Example

Consider:

$$X = (1, 3), \quad Y = (2, 6)$$

From the previous calculations:

$$s_X^2 = 2, \quad s_Y^2 = 8, \quad s_{XY} = 4$$

Therefore, the sample covariance matrix is:

$$S = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix}$$

Interpretation:

- Diagonal values represent variances.
- Off-diagonal values represent covariance.
- Positive covariance shows that X and Y tend to increase together.

Frequentist Statistics

Definition:

Frequentist statistics interprets probability as the long-run frequency of an event after repeated experiments.

Main Idea:

- Parameters are fixed but unknown.
- Probability is estimated using repeated observations.

Equation:

$$P(A) \approx \frac{\text{Number of times event } A \text{ occurs}}{\text{Total number of trials}}$$

Example:

A coin is tossed 100 times and heads occurs 55 times:

$$P(\text{Heads}) = \frac{55}{100} = 0.55$$

Applications: Hypothesis testing, confidence intervals, scientific experiments, and data analysis.

Sampling

Definition:

Sampling is the process of selecting a smaller group, called a **sample**, from a larger group, called a **population**.

Types of Sampling:

- **Random Sampling:** Every member has an equal chance of being selected.
- **Systematic Sampling:** Every k -th member is selected from the population.
- **Stratified Sampling:** The population is divided into groups, and samples are selected from each group.

Application: Sampling saves time, cost, and effort when studying a large population.

Sampling: Example

Example: Student Survey

Suppose a university has 10,000 students, and a researcher randomly selects 500 students for a survey.

Sampling Fraction:

$$f = \frac{n}{N}$$

where:

n = Sample Size, N = Population Size

For this example:

$$f = \frac{500}{10,000} = 0.05 = 5\%$$

Conclusion:

The researcher studies 5% of the population and uses the sample results to make conclusions about the entire population.

IID Sampling

Definition:

IID stands for **Independent and Identically Distributed**. It means that each observation is generated independently from the same probability distribution.

Mathematical Representation:

$$X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} P$$

where P represents the common probability distribution.

Main Properties:

- **Independent:** One observation does not affect another.
- **Identically Distributed:** All observations follow the same probability distribution.

Application:

IID sampling is widely used in statistical estimation and machine learning to analyze data and estimate population parameters.

IID Sampling: Example

Example: Rolling a Fair Die

Suppose a fair die is rolled 5 times:

$$X_1 = 2, \quad X_2 = 5, \quad X_3 = 2, \quad X_4 = 4, \quad X_5 = 2$$

The number 2 appears more than once, which is possible because each roll is independent.

Independent:

The result of one roll does not affect the next roll.

Identically Distributed:

Each roll has the same probability distribution:

$$P(X_i = k) = \frac{1}{6}, \quad k = 1, 2, \dots, 6$$

Therefore:

$$X_1, X_2, \dots, X_5 \stackrel{\text{iid}}{\sim} \text{Uniform}\{1, 2, 3, 4, 5, 6\}$$

Mean Square Error

Definition:

Mean Square Error (MSE) measures the average squared difference between an estimated value and the true value.

Equation:

$$\text{MSE}(\hat{\theta}) = E \left[(\hat{\theta} - \theta)^2 \right]$$

where:

- θ = true parameter
- $\hat{\theta}$ = estimated parameter
- E = expected value

Application:

MSE is used to evaluate the accuracy of estimators and prediction models. A smaller MSE indicates a more accurate estimate.

Mean Square Error: Example

Suppose the true values and estimated values are:

True Values: 2, 4, 6

Estimated Values: 3, 5, 5

Step 1: Find the squared errors

$$(2 - 3)^2 = 1, \quad (4 - 5)^2 = 1, \quad (6 - 5)^2 = 1$$

Step 2: Calculate MSE

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{MSE} = \frac{1 + 1 + 1}{3} = 1$$

Result: The Mean Square Error is 1.

Consistency

Definition:

An estimator is called **consistent** if its estimated value gets closer to the true parameter as the sample size increases.

Mathematical Condition:

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

This means that the probability of the estimator being close to the true parameter approaches 1 as the sample size increases.

Simple Idea:

More Data $\implies$ Better Estimate of the True Value

Application:

Consistency is used to evaluate whether a statistical estimator becomes more accurate when more data is collected.

Confidence Intervals

Definition: A confidence interval is a range of values calculated from sample data that is likely to contain the true population parameter.

A $1 - \alpha$ confidence interval I for an unknown parameter $\gamma \in \mathbb{R}$ satisfies:

$$P(\gamma \in I) \geq 1 - \alpha, \quad 0 < \alpha < 1$$

For the Mean of an IID Sequence:

If the variance satisfies:

$$\sigma^2 \leq b^2$$

then a confidence interval is:

$$I_n = \left[\bar{Y}_n - \frac{b}{\sqrt{\alpha n}}, \bar{Y}_n + \frac{b}{\sqrt{\alpha n}} \right]$$

Key Point:

A confidence interval gives a range rather than a single value, allowing us to quantify uncertainty when estimating a parameter from finite data.

Parametric Model Estimation

Definition:

Parametric model estimation assumes that the data follows a model with a fixed number of unknown parameters. These parameters are estimated using the available sample data.

General Form:

$$Y \sim P_{\theta}$$

where:

- Y = observed data
- P_{θ} = probability model
- θ = unknown parameter to be estimated

Example:

For a normal distribution:

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

The unknown parameters μ and σ^2 are estimated from the sample data.

Application: Used when the underlying probability model of the data is known or assumed.

Non-Parametric Model Estimation

Definition:

Non-parametric model estimation does not assume that the data follows a specific probability distribution with a fixed set of parameters.

Instead, the model is estimated directly from the available data.

General Idea:

Data $\longrightarrow$ Estimate the Model

Example:

A histogram can be used to estimate the distribution of data without assuming that the data follows a normal or any other specific distribution.

Application:

Non-parametric methods are useful when the underlying distribution of the data is unknown or difficult to assume.